A3Q1

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Q1

(a)

```
mlb_bs <- read.csv("MLB_Batting_Boxscores_2024-04.csv")
set.seed(341)
s_idx <- sample(1:nrow(mlb_bs), 500)
mlb_samp <- mlb_bs[s_idx, ]
```

```
N <- nrow(mlb_bs)
n <- 500
pi_u <- rep(n / N, n)

ht_total_bases <- (1 / N) * sum(mlb_samp$total_bases / pi_u)
ht_total_app <- (1 / N) * sum(mlb_samp$plate_appearances / pi_u)

ht_total_bases
```

```
## [1] 1.502
```

```
ht_total_app
```

```
## [1] 3.99
```

(b)

```
estVarHT <- function(y_u, pi_u, pi_uv){
## y_u = an n element array containing the variate values for the sample
## pi_u = an n element array containing the (marginal) inclusion probabilities for the sample
## pi_uv = an nxn matrix containing the joint inclusion probabilities for the sample
delta <- pi_uv - outer(pi_u, pi_u)
estimateVar <- sum( (delta/pi_uv) * outer(y_u/pi_u,y_u/pi_u) )
return(abs(estimateVar))
}
```

```r
pi_uv <- matrix( (n* (n-1)) / (N * (N-1)), nrow = n, ncol = n)

diag(pi_uv) <- pi_u

se_total_bases <- sqrt(estVarHT(mlb_samp$total_bases, pi_u, pi_uv)) / N

se_total_app <- sqrt(estVarHT(mlb_samp$plate_appearances, pi_u, pi_uv)) / N


se_total_bases
```

```
## [1] 0.07747601
```

```r
se_total_app
```

```
## [1] 0.03698137
```

(c)

```r
z <- qnorm(0.75)
ci_total_bases <- c(
  ht_total_bases - z * se_total_bases,
  ht_total_bases + z * se_total_bases
)

ci_total_bases
```

```
## [1] 1.449743 1.554257
```

(d)

```r
set.seed(341)
mlb_top = subset(mlb_bs, batting_order <= 3)
top_idx <- sample(1:nrow(mlb_top), 500)
top_samp <- mlb_top[top_idx, ]
mlb_bot = subset(mlb_bs, batting_order >= 7)
bot_idx <- sample(1:nrow(mlb_bot), 500)
bot_samp <- mlb_bot[bot_idx, ]
```

(e)

```r
N_top <- nrow(mlb_top)
N_bot <- nrow(mlb_bot)

pi_u_top <- rep(500 / N_top, 500)
pi_u_bot <- rep(500 / N_bot, 500)

ht_plate_top <- sum(top_samp$plate_appearances / pi_u_top) / N_top

ht_plate_bot <- sum(bot_samp$plate_appearances / pi_u_bot) / N_bot

ht_plate_top
```

```
## [1] 4.446
```

```r
ht_plate_bot
```

```
## [1] 3.61
```

(f)

```r
# joint inclusion probability

pi_uv_top <- matrix( (500 * 499)/(N_top * (N_top - 1)), 500, 500)
diag(pi_uv_top) <- pi_u_top

pi_uv_bot <- matrix( (500 * 499)/(N_bot * (N_bot - 1)), 500, 500)
diag(pi_uv_bot) <- pi_u_bot


# se
se_plate_top <- sqrt(estVarHT(top_samp$plate_appearances, pi_u_top, pi_uv_top)) / N_top

se_plate_bot <- sqrt(estVarHT(top_samp$plate_appearances, pi_u_bot, pi_uv_bot)) / N_bot

# CI
ci_plate_top <- c(ht_plate_top - z * se_plate_top, ht_plate_top + z * se_plate_top)

ci_plate_bot <- c(ht_plate_bot - z * se_plate_bot, ht_plate_bot + z * se_plate_bot)

ci_plate_top
```

```
## [1] 4.427149 4.464851
```

```r
ci_plate_bot
```

```
## [1] 3.591149 3.628851
```